

1 The Earth may be considered as a uniform sphere of radius $6.37 \times 10^6\text{m}$.

Cambridge is at a point on the Earth’s surface that has a latitude of 52.2° north of the Equator, as shown in Fig. 1.1.

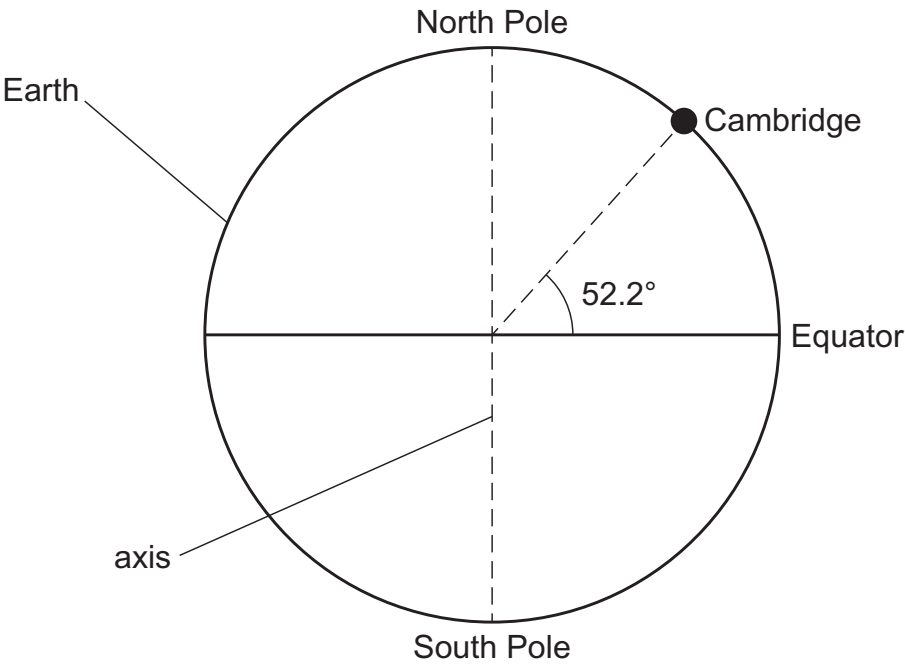


Fig. 1.1

As the Earth spins on its axis, Cambridge moves in a circle that is parallel to the Equator but with a smaller radius.

(a) (i) Show that the radius of the circle around which Cambridge moves is $3.90 \times 10^6\text{m}$.

[1]

(ii) Calculate the speed at which Cambridge moves around the circle.

speed = ms^{-1} [3]

(b) A student of mass 58.6kg stands on horizontal ground in Cambridge.

(i) Determine the magnitude of the resultant force that acts to cause the circular motion of the student.

resultant force = N [2]

(ii) On Fig. 1.2, draw an arrow to show the direction of the resultant force that acts on the student.

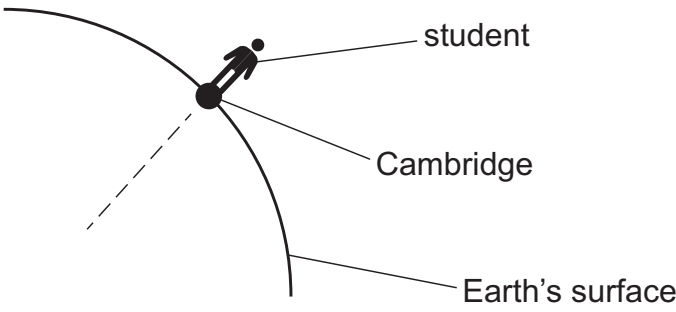


Fig. 1.2 (not to scale)

[1]

(iii) On Fig. 1.3, draw labelled arrows from the student to show the directions of the forces that act on the student to cause the resultant force in (b)(ii).

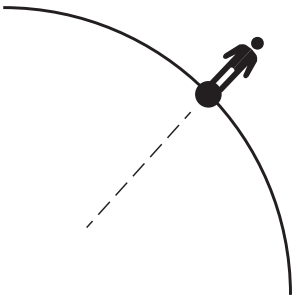


Fig. 1.3 (not to scale)

[2]